

AM 463: QUIZ 3
September 26, in class
15 minutes: Closed Book
Answer all problems
Presentation will affect your grade

Question 1:

Consider the evolution of a tracer in two dimensions (the x-z plane for simplicity) so that

$$\frac{Df}{Dt} = 0.$$

If there is a background flow given by $(0, W(x))$ find the linearized governing equation for tracer perturbations if we write the total velocity as

$$(u, w) = (0, W(x)) + \epsilon(u, w)$$

and the total concentration as

$$f(x, z, t) = \epsilon g(x, z, t).$$

Question 2:

Consider the model equation (it is called the viscous KdV equation, though that doesn't matter for this question)

$$A_t = -cA_x + \alpha AA_x + \beta A_{xxx} + \nu A_{xx}.$$

As in the notes split $A(x, t)$ into an ensemble mean and a fluctuation, $A = \langle A \rangle + a$ so that $\langle a \rangle = 0$. Derive the equation that governs the mean and explain why this equation is not closed. Hint: Recall that $AA_x = (A^2/2)_x$.